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Environment-Based Life Cycle Decomposition (eLCD): Adaptation of EBD to Sustainable Design

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Abstract As sustainability becomes increasingly important, product design is taking a proactive role in producing products that are both useful and sustainable. This paper introduces and discusses a tool named Environment-based life cycle decomposition (eLCD) to adapt the Environment-based Design (EBD) methodology to sustainable design. The eLCD brings to EBD three major features: 1) a holistic environment structure for sustainable conceptual design, 2) an effective and efficient tool for collecting information for sustainability decision-making, and 3) an analysis tool that takes sustainability as an integral part of the design rather than as a burden. The environment of a product is everything except the product itself, which can be defined in three dimensions, namely, environment types, life cycle events, and life cycle time. The environment types are designated as natural, built (including physical artifact and digital artifact), economic, and social environment. The eLCD provides an effective template for information collection to support the design decision-making process. The effectiveness of eLCD is demonstrated by its application to the upscaling of a wind turbine, where an energy storage system is introduced to make full use of wind energy with the least waste in serving the electricity demand.

Keywords: Life cycle decomposition, Sustainable design, Environment-based design (EBD), Wind turbine

Definitions

[1] **Environment-based life cycle decomposition (eLCD)**: a sustainable design tool to model and decompose the environment of a product by analyzing the life cycle of the product considering the product usefulness and sustainability.
[2] **Environment-based design (EBD)**: an environment-centred design methodology that always focuses on the environment of the product to be designed. Any product already designed is a part of the environment for the proceeding design. The environment includes natural, built, and human components. The methodology consists of three activities: environment analysis, conflict identification, and solution generation (Zeng 2011, 2020b).
[3] **Life cycle**: consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal (ISO/TC 2004).

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[4] **Sustainable design**: an approach that puts the well-being of people and the sustainability of the environment first (McLennan 2004), including environmentally sustainable design, socially sustainable design, and economically sustainable design.

1. Introduction

It has been estimated that 80-90 percent of a product's environmental impacts are determined early in the product development process, particularly the conceptual design stage (Hauschild *et al.* 2005). As a result, sustainable design is proposed to avoid the product's unwanted environmental behaviours in its early design stages (Lilley 2009). However, the early design stage does not provide the necessary product details for an effective and feasible environmental assessment (Scrucca *et al.* 2020). Furthermore, the consideration and realization of sustainability will add cost and cycle time for product development; sustainability could be a burden for many companies from the pure business point of view (Baker & Lepech 2009). As sustainability becomes mandatory, companies must be enabled to better balance the interests of all stakeholders during the product development process (Austin-Breneman 2014; Maccioni *et al.* 2019).

Traditional ecological interactions between industrial activities and the environment focused on two-time aspects: the past and the present. Brundtland proposed three sustainability dimensions for the future: economic development, social development, and environmental protection (Brundtland 1987; Gbededo *et al.* 2018; Simpson & Radford 2014). Sustainability is appreciated when the interdependencies of the three sustainability dimensions are considered and analyzed to support decision-making (Gbededo *et al.* 2018; Parent *et al.* 2013; Zamagni *et al.* 2013). Sustainable production has evolved to a philosophy of "cradle to cradle (C2C)" from a traditional production philosophy of "cradle to grave (C2G)" (Dahiya *et al.* 2020; Li 2018). In the meantime, the products need to be environmentally benign and safe during their use (Nathan & Scobell 2012).

The purpose of sustainable design is to produce a product that is useful and sustainable for its working environment (Chen & Zeng 2006). The usefulness of a product can be manifested through its functionality, affordances, and utilities. In achieving the goal, function-based design methodology assumes that a product is designed to satisfy a set of prescribed functions, which may imply a functional structure for a given design problem (Gero 1990; Goel *et al.* 2009; Julie *et al.* 2002; Pahl & Beitz 2013; Umeda *et al.* 1990). Design solutions can usually be abducted from a functional structure (March 1976; Roozenburg 1992). Along the same line, affordance has been introduced to represent how a product can interact with and serve its working environment (Maier & Fadel 2009). Affordance provides wider coverage than function in terms of describing a product's usefulness. User-Centred Design (UCD) (Norman 1988), Environment-Based Design (EBD) (Zeng 2004, 2015), and Situated Function-Behaviour-State (S-FBS) (Gero & Kannengiesser 2004) are alternative methodologies that start a design process from the environment (user in case of UCD, all environment in case of EBD, and the context in case of S-FBS) of a product. A comprehensive review of how to achieve the usefulness of a product can be found in the literature (Wynn & Clarkson 2018). Existing design methodologies have yet to be well integrated with sustainability. This paper adapted Environment-Based Design (EBD) methodology by proposing an environment-based life cycle decomposition (eLCD) tool to find the key design requirements in sustainable design considering the product usefulness and sustainability.

1.1. Environment-based design (EBD)

Zeng proposed the Environment-Based Design (EBD) methodology to start the design process from the environment (Zeng 2004, 2015; Zeng & Gu 2001). The traditional design methodologies generate product descriptions based on functional or affordance structures, which, however, cannot be well defined before the unknown product structure is generated. This is the basic dilemma implied in the nature of design, which is shown by the recursive logic of design (Zeng & Cheng 1991). Zeng argued that the environment should be the first thing to look at in a conceptual design while the product is the last (Zeng 2020b). As shown in Figure 1, the EBD includes three activities: environment analysis, conflict identification, and solution generation. Design problem, design knowledge, and design solutions simultaneously and interdependently evolve as a part of the environment.

Since the EBD considers all of the environment components relevant to a product design (Zeng 2004, 2020b), it is natural for the EBD to consider sustainability by zooming in on the critical sustainability environment. This can be achieved by customizing the life cycle decomposition in EBD according to the characteristics of sustainability, based on which the EBD design process is also customized. The sustainability-focused environment analysis method is thus called Environment-based life cycle decomposition (eLCD).

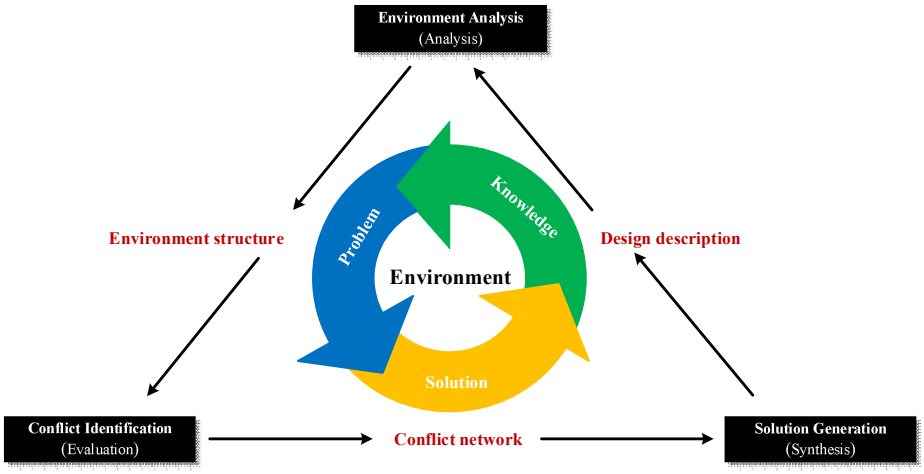


Figure 1 Environment-based Design (EBD)

1.2. Contributions: eLCD is a tool to support environment-based sustainable design

The goal of sustainable design is to help designers make the right decisions to achieve the sustainability of the environment while creating values through design innovation. To make the right decisions, designers must be equipped with the right knowledge; to develop the right knowledge, designers must collect the right information; to collect the right information, designers must ask the right questions; finally, to ask the right questions, designers must understand right the problem. This process is shown in Figure 2.



Figure 2 Decision-making Process in Design

Therefore, to generate effective and efficient sustainable design solutions, the following questions must be answered first:

- Q1: How to understand what is meant by a text describing a design problem?
- Q2: How to ask the right questions based on an understanding of the design problem?
- Q3: How to collect the right information from questions related to the design problem?
- Q4: How to develop design knowledge from the collected information regarding the design problem?
- Q5: How to make the right decisions for the design problem?

While EBD generally addresses all of the five questions above, this paper introduces and discusses a tool named Environment-based life cycle decomposition (eLCD), facilitating the application of EBD to sustainable design. Particularly, the eLCD brings EBD a holistic environment structure for sustainable

design, which is the foundation to address Q3 and Q4. The eLCD could simultaneously engage environmental, economic, and social aspects of sustainability without being limited to specific products or design stages. The analysis is centred around the product environment, which is modelled in three dimensions: environment types, life cycle events, and life cycle time. Four environment types are defined: nature, built (including physical artifact, digital artifact), economic, and social environment.

The rest of the paper is organized as follows: Section 2 introduces the structure of eLCD, including the life cycle of products and environment types. Section 3 presents a case study, which is to generate key sustainable design requirements of a wind turbine by upscaling from a 5-MW to 10-MW, using the eLCD to analyze the life cycle of the wind turbine, which leads to a new wind turbine design in alignment with the requirement of sustainability. Section 4 briefly discusses two basic questions related to knowledge and uncertainty. Finally, section 5 concludes this paper.

2. eLCD-guided Environment Structure

In alignment with a basic statement in EBD methodology that everything except for the product is product environment (Zeng 2002), the eLCD views anything existing, known, and defined at the moment of designing as environment, whereas the unknowns to be created as the product. Extended from the environment structure defined in terms of the product life cycle and environment classification (Chen & Zeng 2006; Zeng 2020b), the environment structure of eLCD can be defined in Figure 3 with three dimensions: environment type, life cycle events, and life cycle time. This structure guides a designer to better understand "what" during the design process at the conceptual stage. The details of each dimension will be addressed in the following subsections.

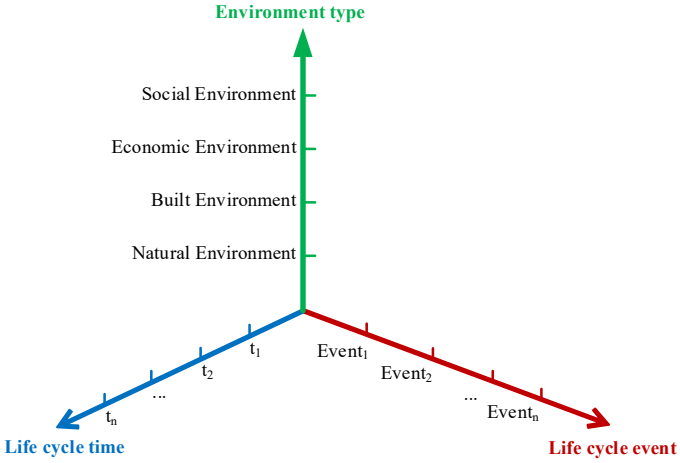


Figure 3 Product Environment Structure for eLCD

2.1. Life cycle time and events

The life cycle is defined as consecutive and interlinked stages of a product system, from the raw material acquisition or generation from natural resources to final disposal (ISO/TC 2004). The life cycle is also defined as a set of interrelated or interacting activities that transform inputs into outputs (ISO/TC 2015b). ISO 14001 divides the life cycle into sequential stages that include acquisition of raw materials, design, production, transportation/delivery, use, end-of-life treatment, and final disposal (ISO/TC 2015a). Chen and Zeng noted that while a product life cycle can generally be divided into various kinds of events, such as design, manufacture, sales, transportation, use, maintenance, and recycle, some events may occur simultaneously (Chen & Zeng 2006). Thus, from the life cycle point of view, environment structure can be defined in a two-dimension space: life cycle time and life cycle events (Zeng 2020b).

While life cycle time is self-evident in that it represents the time points in a product's life cycle, the following defines sustainability-specific life cycle events. Based on the observation and summary from the

literature (Chen & Zeng 2006; ISO/DIS 2015; ISO/TC 2004; Maccioni *et al.* 2019; Williams 2009), eLCD includes eight life cycle events, which are the acquisition of resources, design, manufacture, sales, transportation, use, maintenance, and end-of-life treatment, as defined below:

[Acquisition of resources] The acquisition of resources, which are raw materials and natural resources, aims to get resources from nature as the input to design and manufacture products.

Raw materials include primary or secondary materials that are used to produce a product. Recycled materials are a part of secondary materials (ISO/TC 2004). The raw material acquisition consists of material harvesting and transportation to manufacturing sites (Williams 2009). Natural resources include sunlight, wind, atmosphere, water, oil, land (including all minerals), forests, plants, and animals.

[Design] Design is a recursive process to create novel and useful products that serve and change its working environment (Zeng 2015; Zeng & Cheng 1991; Zeng & Gu 1999).

In this event, designers are faced with a design problem or a situation, analyze it, and generate design solutions. Sustainable design must make sure that the change of environment will consume the minimum natural resources and cause minimum or no damage to the natural environment. Details will be discussed in Section 0.

[Manufacture] Manufacture is the realization of a design solution into a real product. In this event, the input is the raw materials, and the output is the finished product.

During the process, many important factors need to be considered, such as safety, economics, and remaining waste which affects the environment.

[Sales] Sales are to offer an available product or deliver a service to customers, which are involved with goods/service, marketing, seller/provider, and buyers.

[Transportation] Transportation is the movement of goods from one place to another, which may appear at different life cycle times, such as after the manufacturing, after the recycling, or at the disposal.

Modes of transportation are determined by many factors such as the size of the goods, the type of materials, the distance between two places, and the transportation cost.

[Use] Use of a product depends on how its users perceive its functions, affordances, or usefulness, which is involved with the product, user, and environment. In sustainable design, one needs to pay special attention to the impact of product use on the natural environment, human environment, and social environment.

[Maintenance] Maintenance is the process aimed to keep, repair, or upgrade the product to keep it in good condition. Maintenance can be categorized into three types: preventive maintenance, breakdown maintenance, and preventive maintenance and upgrade. Preventive maintenance happens when the function is degraded. Breakdown maintenance happens when the product has a failure. Preventive maintenance and upgrade happen when the required functional level has changed.

[End-of-life treatment] End-of-life is the point in time when the product no longer satisfies the initial purchaser or first owner (Rose *et al.* 2001). End-of-life treatment can be categorized into two types: remanufacture and disposal. Remanufacture may comprise remanufacture, recondition, repair, reuse, or recycle to end-of-life products (Paterson *et al.* 2017). Disposal may include incineration, landfilling (Rose *et al.* 2001).

Under this context, we could analyze the life cycle of eLCD. There are two events in the environment of eLCD, which are design and monitoring. During the conceptual and detailed design stage, designers need to analyze the environment of the desired product. Then, after designers completed the design, the monitoring organization will assess the product for compliance with certain standards. The related environment types on each life cycle event will be discussed in Section 2.2.

2.2. Environment types

It is an overwhelming challenge for designers to consider all the environment components when they design a product; hence, it is necessary to understand the basic environment types that can be used to represent and rank complex environments. Chen and Zeng proposed three basic types of environment: natural, built, and human, to represent a product's working environment (Chen & Zeng 2006); however, these three types of environment need to be adapted to an application domain for the effective

implementation of EBD. Therefore, in the context of sustainable design, four types of base environment are identified, as shown in Figure 4, which are natural, built, economic, and social environment.

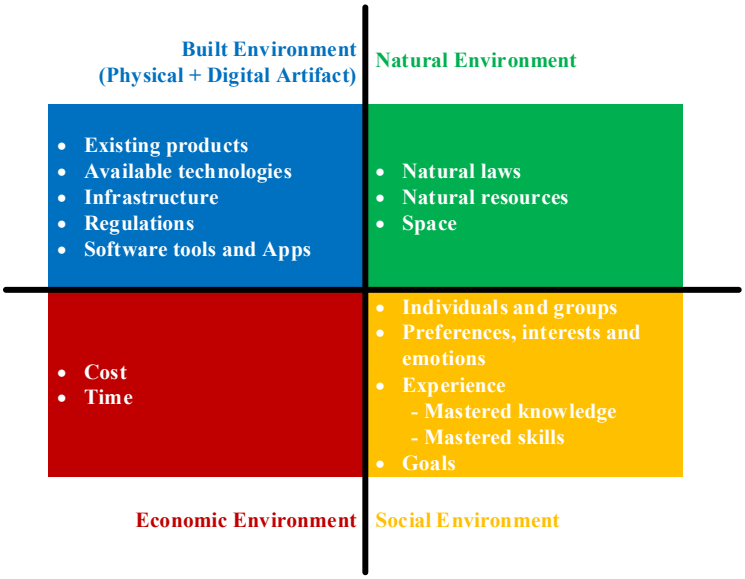


Figure 4 Environment Types Categorized by Requirements

The natural environment is fundamental since all products must be built based on natural laws and rules (Chen & Zeng 2006). The built environment includes physical artifact and digital artifact environments. The physical artifact environment includes existing products, available technologies, and infrastructure. The digital artifact environment embodies documents, data sets, methodologies, software tools and apps, and regulations such as social laws, rules, norms, and technical regulations. The economic environment includes financial resources such as cost and time. The social environment includes individual and group stakeholders (Getahun & Selassie 2017) and their goals, experience, preferences, interests, and emotions. Stakeholders could change their goals and interests interdependently over time (Getahun & Selassie 2017; Merrilees *et al.* 2005). Effective social engagement is fundamental to ensure the sustainability of the earth ecosystem in the entire life cycle of a product system.

Figure 4 provides a template for eLCD to collect comprehensive information relevant to the sustainable design of a product. The template will be used in the question-driven process to be introduced in Section 0, which will define different environment components according to the four types adaptive to different products.

3. Case Study

In this section, a case study to upscale 5 MW wind turbine to 10 MW wind turbine is presented to apply eLCD to a wind turbine design in sustainable design. The existing 5 MW wind turbine has a structure shown in Figure 5.



Figure 5 5 MW Wind Turbine Structure

With this structure, the wind turbine will produce power only when the wind speed falls between cut-in and cut-out speed, as shown in Figure 6. In addition, when there is no contracted sale to the electricity grid, the wind turbine will also be shut down.

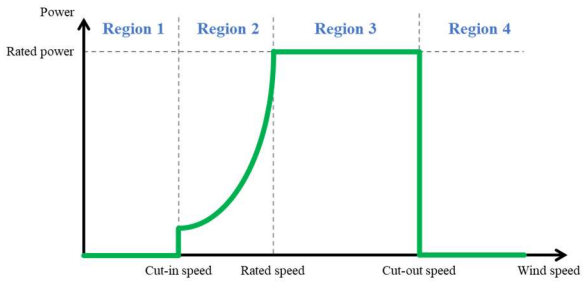


Figure 6 Power Extracted by a Wind Turbine (adapted from (Aho *et al.* 2013)**)**

To make the wind energy competitive with other forms of electricity, the cost of wind energy must be lowered as much as possible. The cost of wind energy can be defined as the costs of finance, construction, installation, and maintenance and divided by the energy yield (Van Kuik *et al.* 2016). Three fundamental drivers have played a role in reducing the cost of wind energy, namely, increased hub height, power rating, and rotor diameter (Veers *et al.* 2019). Increased hub height allows access to higher-quality wind resources (i.e., increase free-stream air velocity and reduced surface friction effects). Increased rotor diameter enables a greater capture of the energy by larger wind turbine blades. The increased hub height and larger blades are now the main driver and focus for upscaling wind turbines.

There are many traditional upscaling designs for wind turbines. Theoretical upscaling (Sieros *et al.* 2012) is based on self-similar geometric upscaling while assuming geometric and aerodynamic similarity. Among others, this method does not capture wind shear effect or size effects related to the fracture mechanics. Real upscaling (Sieros *et al.* 2012) is based on observed industrial trends—hence actual loads scale use in available designs, weight increase law, and cost increase law. Optimum upscaling (Sieros *et al.* 2012) is based on the wind turbine life cycle in terms of costs and income. System-level design (Zalkind *et al.* 2019) focuses on developing a simplified model that yields a relationship between structural loads in the wind turbine. The simplified model is then used to explore the design space. Multidisciplinary design optimization (Ashuri *et al.* 2016) involves using scales to develop an initial optimized wind turbine. Using this initial optimized design, the relation between design parameters and rotor diameter is extracted and used to calculate more accurate trends.

A new conceptual structure of wind turbine is designed to produce 10 MW electricity output whenever is needed, as shown in Figure 7, where an energy storage system is added to decouple the rotor from the gearbox. With this design, the rotor, within its capacity, will capture all available wind energy to store in the energy storage system, which can be realized with different mechanisms, such as spring. When and only when there is a demand from the electricity grid, the energy storage system will be turned on to run the gearbox, further driving the generator to produce the needed electricity.

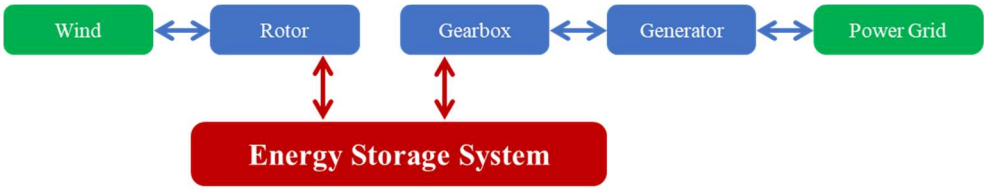


Figure 7 A Structure of 10 MW Wind Turbine with the Concept of a Kinetic Energy Storage System

This new design in Figure 7 indeed realizes the requirement of sustainable design, which combines product usefulness and sustainability for wind energy. The new wind turbine boosts wind energy resource effectiveness and efficiency through the kinetic energy storage system. This energy management system can also well support the pay-per-use principle for the electricity grid. While the electricity market informs the wind turbine design, the wind farm provides efficient value creation through innovative designs of new

wind turbines. This naturally produced sustainable wind turbine design shows the effectiveness of the proposed eLCD. The detailed mechanism of the design solution will be reported in a separate paper. The input of eLCD is a design statement, and the output is a key design requirement for wind turbine upscaling. In the following, we will detail each step of eLCD supported EBD process, as specified in Appendix A, for the wind turbine sustainable design.

3.1. Understanding a design problem

The design statement is "upscale 5 MW wind turbine to 10 MW wind turbine". The objective is to make better use of wind resources. Designers could draw a ROM diagram for this statement based on EBD, as shown in Figure 8.

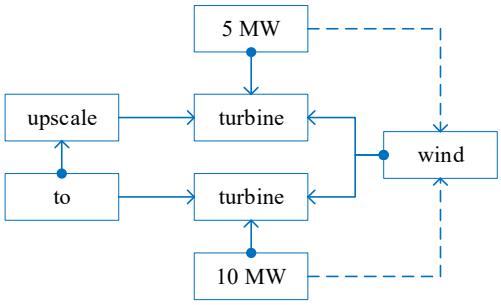


Figure 8 ROM Diagram of Wind Turbine Design

3.2. Decomposing life cycle

By applying the two rules in Table 7 to the ROM diagram in Figure 8, it can be determined that the object 'wind' should be asked a question first. Using the question template in Table 8, the question "what is the wind?" should be asked. Table 1 shows the order of questions thus generated, which are used to clarify and refine the wind turbine upscaling problem.

Table 1 Questions to Clarify the Problems for Wind Turbine Design

Questions
Q1: What is the wind?
Q2: What is a wind turbine?
Q3: What is the 5 MW wind turbine?
Q4: What is the 10 MW wind turbine?
Q5: Why upscale the 5 MW wind turbine to 10 MW wind turbine?
Q6: Who to upscale the 5 MW wind turbine to 10 MW wind turbine?
Q7: How to upscale the 5 MW wind turbine to 10 MW wind turbine?

Table 2 lists the answers to the questions in Table 1, following the guideline given in Table 9. It must be noted that in the actual design process, the answers do not need to be given for all the questions in one iteration. Three approaches can be applied, which are depth-first, breadth-first, and hybrid (Wang *et al.* 2015; Zhao & Zeng 2019). It was found that experts usually take a hybrid approach while novices usually take a depth-first approach (Atman *et al.* 2007). The breadth-first approach is taken in this case study simply for the sake of convenience of presentation.

While it is clear that answers to Q2-4 followed the environment structure defined by Figure 3, Figure 4, and the answer to Q1 may not be so obvious. Indeed, the "movement" is the wind's life cycle event, "time" is the wind's life cycle time, and "space" is a wind's environment type. After the answers to Q2-4 are merged into the ROM diagram in Figure 8, the same questioning and answering process will lead to Q8 and Q9, as shown in Table 3.

Table 2 Answers to Questions in Table 1

Q1: What is the wind?

The wind is the movement of air from high-pressure to low-pressure areas, which is usually caused by the differences in temperatures between two surfaces due to solar energy absorption. The wind speed and direction vary in time and spatial scales (Burton *et al.* 2003; Walker & Swift 2015).

Q2: What is a wind turbine?

A wind turbine is a device that converts the wind's kinetic energy into electrical energy.

Q3: What is the 5 MW wind turbine?

A 5 MW turbine is a wind turbine that produces 5 MW of electrical power at rated power conditions.

The lifecycle of a wind turbine includes the acquisition of resources, design, manufacture, transportation, erection, use, maintenance, and end-of-life treatment.

For the *event of acquisition* of resource, environment components are listed as follows:

a) the **natural environment** includes natural ground in which there are many kinds of raw materials or resources such as steel, glass fibre, epoxy resin, copper, aluminum (Garrett & Rønde 2013a).
b) the **built environment** includes:
    i. the physical artifact environment includes
    - extraction and processing technologies of materials,
    - origin countries of materials,
    - supply chains whose location decides the cost and convenience of transportation of wind turbines,
    - remaining waste which might affect global warming, marine aquatic ecotoxicity, and freshwater aquatic ecotoxicity (Garrett & Rønde 2013a).

    ii. the digital artifact environment includes laws related to resources.
c) the **economic environment** includes all costs and time.
d) the **social environment** includes technicians, acquisition businesses, raw materials sellers, governments.

For the *event of design*, environment components are listed as follows:

a) the **natural environment** comprises offshore or onshore, with a wind classification of high, medium, and low.
b) the **built environment** includes:
    i. the physical artifact environment includes a physical structure that has four main components, namely, rotor, drivetrain, tower, and foundation.
    ii. the digital artifact environment includes technical resources and design tools.
c) the **social environment** includes a multidisciplinary team in which designers have expertise in
    - aerodynamics, flow physics, and hydrodynamics, in the case of offshore applications,
    - structures, geomechanics, and material science,
    - dynamics and controls, and
    - power flow, as well as power electronics (Van Kuik *et al.* 2016).

For the *event of manufacture*, environment components are listed as follows:

a) the **natural environment** includes (Garrett & Rønde 2013b; Gkantou *et al.* 2020)
    - steel (used more than 80%) for nacelle, tower, and foundation,
    - glass fibre and epoxy (less than 7%) for rotors, and
    - copper and aluminum (less than 2%) for nacelle.
b) the **built environment** includes
    i. the physical environment includes
    - rotor, tower, nacelle, and foundation, (Garrett & Rønde 2013b; Haapala & Prempreeda 2014),
    - processing technologies from materials for a wind turbine,
    - all the cost, and
    - remaining waste which might affect global warming, marine aquatic ecotoxicity, and freshwater aquatic ecotoxicity (Garrett & Rønde 2013a), which have to be assessed in terms of their main parts and sub-assemblies, rather than the entire system (Haapala & Prempreeda 2014).

ii. the digital environment includes electricity data which can reflect the accuracy of wind turbines.

c) the **economic environment** includes all costs and time.
d) the **social environment** includes technicians, manufacturing facilities, wind farms (Garrett & Rønde 2013b; Haapala & Prempreeda 2014).

For the *event of transportation*, environment components are listed as follows:

a) the **natural environment** includes
- fossil fuels,
- weather, high gales or storms may prevent the transportation and installation of the components to the site, which results in delays of wind turbines transportation that increases the cost of the project,
- distances between wind farms and manufacturing facilities are usually between 20 to 50 kilometres (Haapala & Prempreeda 2014).

b) the **built environment** includes

i. the physical artifact environment includes:
- various vehicles such as ships, lorries, and trains or their combination thereof depending on the location of the suppliers (Haapala & Prempreeda 2014),
- vehicle fuels combustion on the way between the wind farm and manufacture or incinerated areas,
- components of the production, and

ii. the digital artifact environment includes
- size laws which limit the size of transported wind turbine objects, and
- special permits of different governments.

c) the **economic environment** includes all costs and time.
d) the **social environment** includes drivers, wind farms operators, manufacturing facilities, governments.

For the *event of erection*, environment components are listed as follows:

a) the **natural environment** includes fossil fuels,
b) the **built environment** includes

i. the physical artifact environment includes site cables, transformer station, existing grid cranes, diggers, other vehicles,
ii. the digital artifact environment can be ignored.

c) the **economic environment** includes all costs and time.
d) the **social environment** includes technicians, wind farms.

For the *event of use and maintenance*, environment components are listed as follows:

a) the **natural environment** includes:
- wind (low, medium, high), and
- animals, such as birds, that appear in the environment.

b) the **built environment** includes

i. the physical artifact environment includes:
- characteristics of a wind turbine such as the number of wind turbines per area, height of the turbine, blade size,
- electricity generation and transmission,
- remaining waste which influences terrestrial ecotoxicity potential, human toxicity potential; abiotic resource depletion (elements), and
- wind turbines noise influences the health of local people close to the wind turbine (Garrett & Rønde 2013a).

ii. the digital artifact environment includes wind-farm control software such as SCADA (Walker & Swift 2015).

c) the **economic environment** includes all costs and time.
d) the **social environment** includes technicians, wind farms, electric facilities, customers.

For the *event of end-of-life treatment*, environment components are listed as follows:

a) the **natural environment** includes
- fossil fuels,

- weather, high gales or storms may prevent the transportation and installation of the components to the site, which results in delays of wind turbines transportation that increases the cost of the project,
- distance between the wind farm and the manufactures or incinerated areas.

b) the **built environment** includes

i. the physical artifact environment includes:

- various vehicles such as ships, lorries, and trains or their combination thereof depending on the location of the suppliers (Haapala & Prempreeda 2014),
- recyclable materials (steel, copper, aluminum, and cast iron), non-recyclable materials (concrete, fibreglass), which can reduce the impact on global warming or marine aquatic ecotoxicity potential (Garrett & Rønde 2013a).

ii. the digital artifact environment includes

- size laws which limit the size of transported wind turbine objects, and
- special permits of different governments.

c) the **economic environment** includes all cost and time.

d) the **social environment** includes drivers and policemen, wind farms, manufacturing facilities, governments.

Q4: What is the 10 MW wind turbine?

It is a wind turbine that produces 10 MW of electrical power at rated power conditions.

Q5: Why upscale the 5 MW wind turbine to a 10 MW wind turbine?

Wind farms need to increase the production of wind power at a minimum cost in response to dynamic demands from the electricity grid. (Ashuri *et al.* 2016; Sieros *et al.* 2012; Zalkind *et al.* 2019).

Q6: Who to upscale the 5 MW wind turbine to 10 MW wind turbine?

Designers.

Q7: How to upscale the 5 MW wind turbine to a 10 MW wind turbine?

The question can be ignored at this stage because the answer is the final product, which should be looked at last.

Figure 9 to Figure 12 show the updated ROM diagram after answers in Table 2 and Table 3 are merged into the initial ROM diagram in Figure 8. It is noted that due to the limit of the space, we do not include transportation and end-of-life treatment events in answers to Q3; instead, we will focus only on the acquisition of resource, design, use, and maintenance, which are more relevant to the current design. This does not affect the validity of the proposed methodology.

Table 3 Iterative Questioning and Answering from Answers to Q2-4 in Table 2

Q8: What is the wind's kinetic energy (E_w)?

Wind's kinetic energy is the kinetic energy contained in the air flow, which is characterized by the following equation:

$$E_w = P_w t = \frac{1}{2}\rho A U^3 t$$

where P_w is wind power (kW), time (t) is in hours (h), ρ is the air density, *A is r*otor swept area ($A = \pi R^2$ (m^2)), and U is wind speed.

Q9: What is electrical energy (W)?

Electrical energy is the electrical power per unit of time produced by a wind turbine in which the turbine losses aerodynamic power in transmission and generator:

$$W = Pt = C_P \eta_M \eta_G P_w t$$

where P_w is wind power (kW), time (t) is in hours (h), C_P is power coefficient, η_M is efficiency, η_G is generator efficiency.

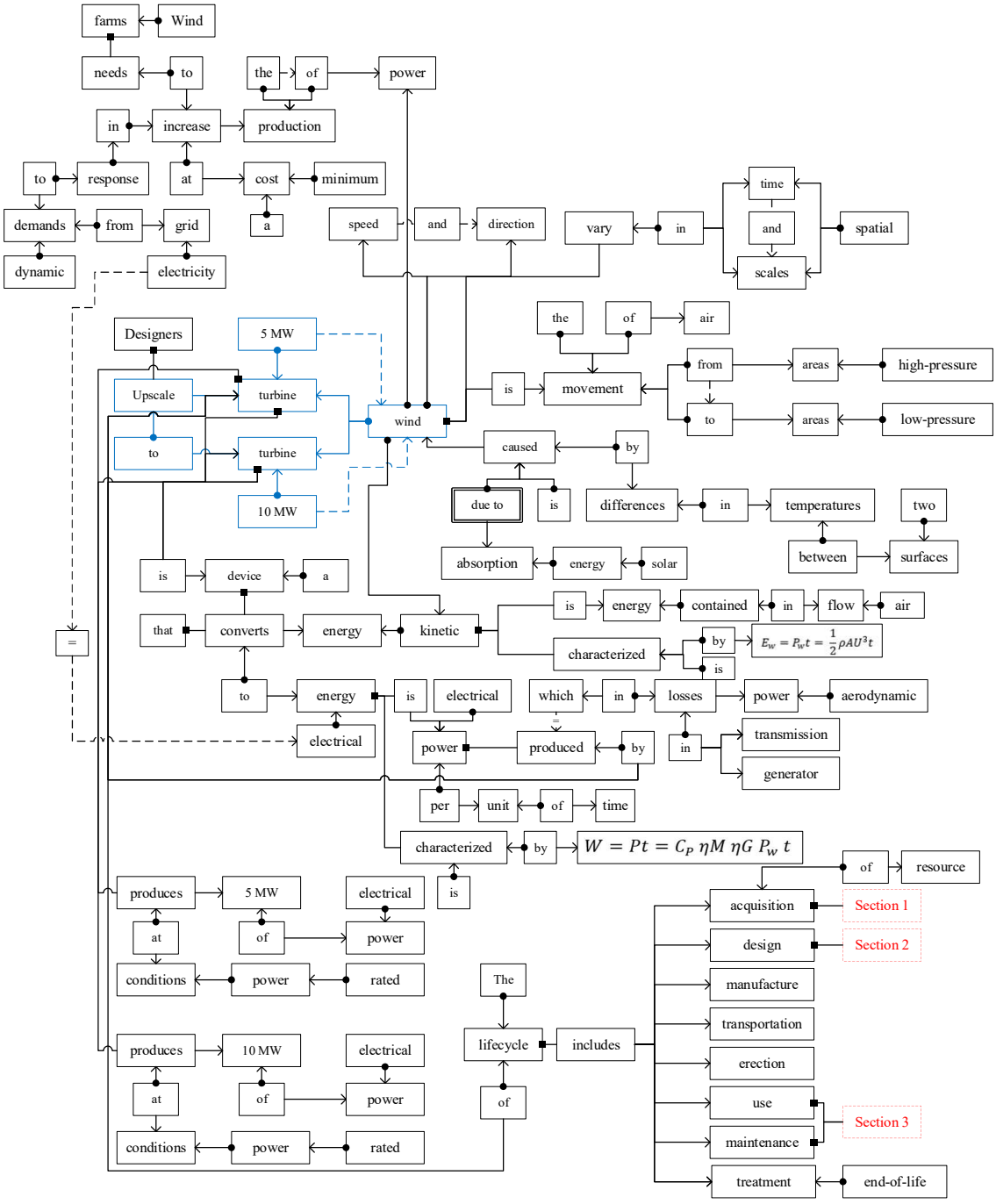


Figure 9 ROM Diagram Updated by the Answers

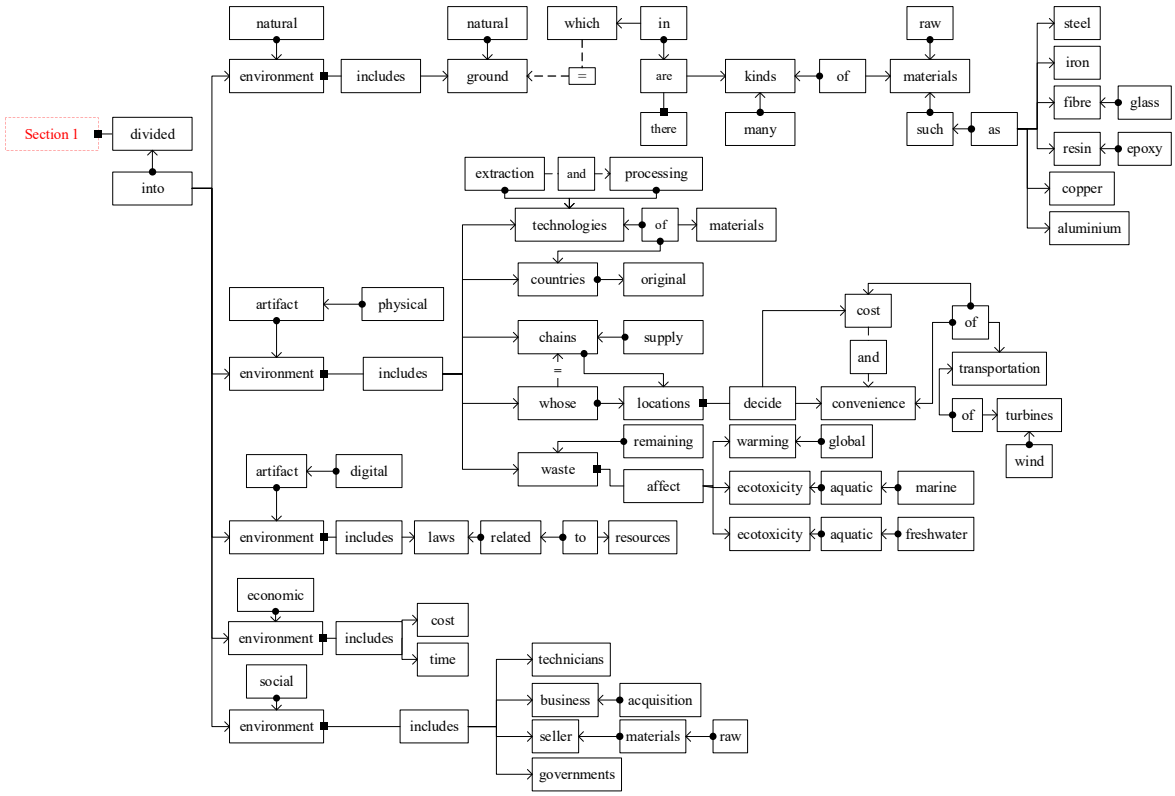


Figure 10 Partial Section 1 ROM Diagram Updated by the Answer of Q3

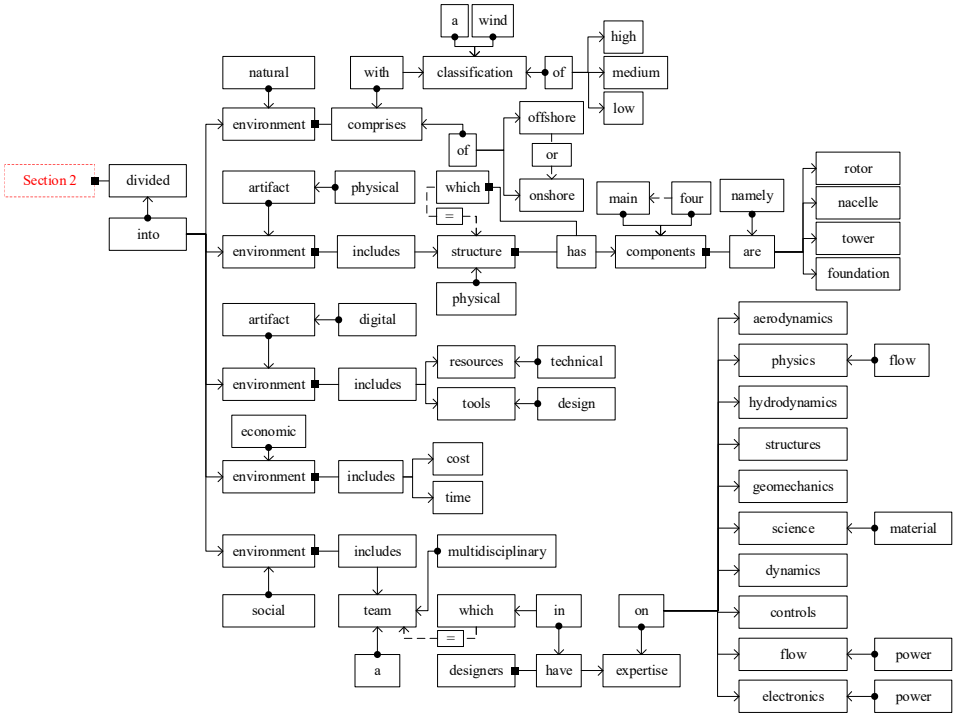


Figure 11 Partial Section 2 ROM Diagram Updated by the Answer of Q3

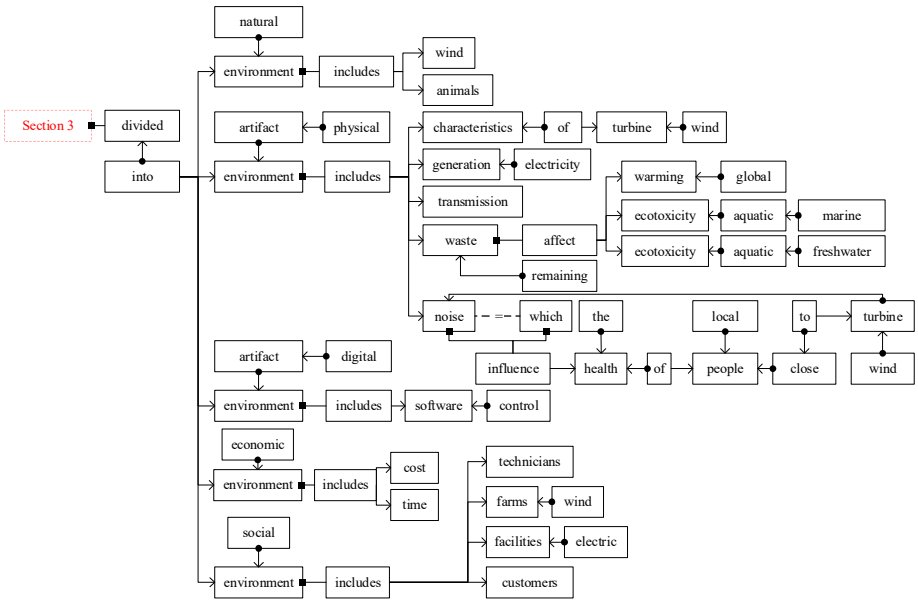


Figure 12 Partial Section 3 ROM Diagram Updated by the Answer of Q3

3.3. Modelling life cycle dependency

This step is to find explicit and implicit interactions from the updated ROM diagram and to construct the interaction dependency network. The interactions from the ROM diagram in Figure 9 are listed in Table 4.

Table 4 Interactions for Wind Turbine Design

Interaction#	*Description*
I1	The wind speed and direction vary in time and spatial scales.
I2	Differences in temperatures between two surfaces cause the wind.
I3	Wind farms need to increase the production of wind power at a minimum cost in response to dynamic demands from the electricity grid.
I4	Designers upscale 5 MW wind turbine to 10 MW wind turbine.
I5	Air flow contains the wind's kinetic energy.
I6	A wind turbine produces electrical energy.
I7	The wind turbine losses aerodynamic power in transmission and generator.
I8	The 5 MW wind turbine produces 5 MW of electrical power at rated power conditions from the wind.
I9	The 10 MW wind turbine produces 10 MW of electrical power at rated power conditions from the wind.
I10	The remaining waste affects global warming, marine aquatic ecotoxicity, and freshwater aquatic ecotoxicity.
I11	Wind turbine noise influences the health of people close to the wind turbine.

The interaction dependence network is shown in Figure 13, which is constructed using the definition of interaction dependence relation shown in Figure 20.

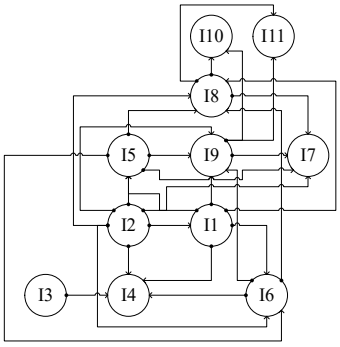


Figure 13 Interaction Dependency Network for Wind Turbine Design

Figure 14 shows the simplification of the network in Figure 13 by removing a dependence relation between two interactions if there is a longer dependence path between the two. In Figure 14, two specific interactions are singled out: independent interactions and dependence-only interactions. Independent interactions (indicated in red) are outgoing interactions that are not dependent on any other interactions, such as I2 and I3. Dependence-only interactions (indicated in blue) do not constrain any other interactions such as I4, I7, I10, and I11. All the other interactions are translational that forms different sub-design problems by connecting independent interactions and dependence-only interactions.

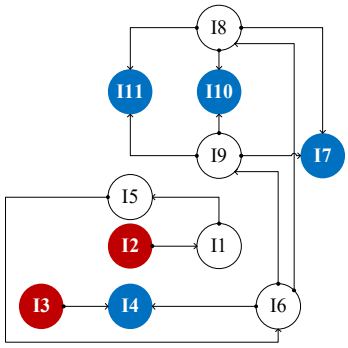


Figure 14 Simplification of Interaction Dependence Network (Independent Interactions in Red and Dependency-Only Interaction in Blue)

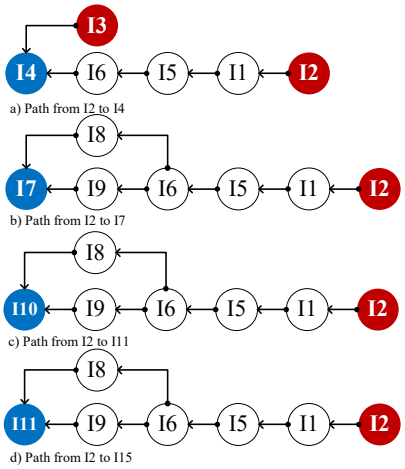


Figure 15 Process Paths Included in the Interaction Dependence Network (Independent Interactions in Red and Dependency-Only Interaction in Blue)

All the potential paths leading to a dependence-only interaction from independent interactions are shown in Figure 15. I2 and I3 are necessary for I4, whereas I2 is also necessary for I7, I10, and I11.

Since I4 is the major concern of the design at this stage, we will focus on I4 in this case study. The same procedure can be applied to I7, I10, and I11. I2 and I3 are key interactions that constrain I4, while I1, I5, and I6 are translational interactions critical for the implementation of I4, as shown in Table 5.

Table 5 Key and Critical Interactions for Implementing Wind Turbine Upscaling

	Interaction#	Description
Key	I2	Differences in temperatures between two surfaces cause the wind.
	I3	Wind farms need to increase the production of wind power at a minimum cost in response to dynamic demands from the electricity grid.
Translational	I1	The wind speed and direction vary in time and spatial scales.
	I5	Air flow contains the wind's kinetic energy.
	I6	A wind turbine produces electrical energy.

The discussion above reveals the major problem in the current wind turbine design: the conflict between the varying wind velocity and the dynamic demands from the electricity grid. It is this conflict that led to the design ideation shown in Figure 7.

4. Discussions

Knowledge is an integral part of the sustainable design process. Since the logic of design is recursive (Zeng & Cheng 1991), which implies a nonlinear dynamic in that design problem, design knowledge, and design solutions interdependently change each other in the design process (Nguyen & Zeng 2012), it is challenging to determine which knowledge can be used at each major stage of the design. This uncertainty in determining knowledge is an inevitable challenge for designers in the design process (Baker & Lepech 2009).

Uncertainty results from either a lack of the necessary knowledge for the design or a wrong application of an existing piece of knowledge to a given design problem. In the case of the lack of knowledge, the design problem remains to be open, and no proper solution can be found, which leads to risk and failure. Therefore, designers need to collect and integrate related information and knowledge. However, many designers, particularly novice designers, do not know what, when, and how to collect and integrate knowledge. Designers could search and study knowledge through the ROM, question template, and the answering guideline in the eLCD. When designers explore the life cycle of environment components, they are acquiring knowledge about the environment. Furthermore, the prerequisites for eLCD are light for a designer since domain knowledge and experience are the results rather than the premise for applying eLCD.

If knowledge cannot be properly applied to the right context, then fixation will happen (Nguyen & Zeng 2017). There are three paths to design fixation: 1) unable to abandon potential design solutions or to make major potential design solutions changes to design solutions, 2) do not have the right perception of the current requirement, and 3) do not properly execute due to poor evaluation of solution-requirement. In the case of too much knowledge, the new design problem and its potential solution are the same as the old design problem, which leads to failure. Based on the process of eLCD, designers could get the key activity through correctly understanding a design requirement. Along with the arrow direction in the ROM diagram, designers could identify all explicit interactions. During this process, designers might know what they should do or should not do about decision-making. For example, designers might understand what interaction is important and urgent, important and unurgent, unimportant and urgent, and unimportant and unurgent based on the interaction dependency network, which could help them make a next step plan about

the design task. Besides, if designers would like to redesign a product, an interaction dependency network might also help them find the important part to redesign.

5. Conclusion and Future Work

In this paper, a new life cycle decomposition tool based on the basic principles and concepts of Environment-based Design (EBD) design methodology, called Environment-based life cycle decomposition (eLCD), is proposed for sustainable conceptual design. There are three characteristics of eLCD: 1) a holistic and sustainable tool that contains natural environment, economy, and social dimensions and also contains built environment including physical artifact, digital artifact environment, 2) an effective and efficient tool for collecting information for sustainability decision-making, and 3) an analysis tool that takes sustainability as an integral part of the design rather than as a burden. The case study of the upscaling of wind turbines, followed by three eLCD activities, shows that the proposed eLCD model is feasible and applicable in the practical design process and could improve designers' creativity to get a creative design solution.

Research shows that people would like to focus on design solutions first when given a design task though looking at the environment first can be more beneficial. However, many designers do not like analyzing the environment since this does not lead to a solution right away. The need for eLCD to conduct the environment decomposition before a solution is produced could generate an uncomfortable feeling for some designers. It is thus important, as future work, to study how to balance the workload in the eLCD approach so that it can be more designer friendly.

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Appendix

A. eLCD-guided Problem Formulation of Sustainable Design

The eLCD is the customization and adaptation of EBD to sustainable design with an emphasis on environment analysis. The objective of eLCD is to identify known environment components and their cause-effect relationships to achieve sustainable design effectively. The input of eLCD is a design problem statement which is usually described by a natural language. In contrast, the output is an interaction dependence network that captures cause-effect relationships among activities between environment components. Under this context and corresponding to the general decision-making process illustrated in Figure 2, the eLCD strategy is a question-driven process.

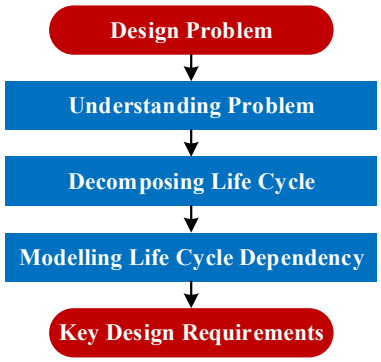


Figure 16 Process of Environment-based Life Cycle Decomposition (eLCD)

The eLCD can be divided into three steps: understanding a design problem, decomposing life cycle, and modelling life cycle dependency, as shown in Figure 16. Firstly, based on the recursive object model (ROM), understanding a design problem aims to help designers capture the critical messages implied in a problem description (Zeng 2008). Secondly, decomposing life cycle, based on an EBD strategy for asking the right questions related to the design and the eLCD structure introduced in Section 2, aims to collect the necessary and sufficient information about the environment components and their relationship for sustainable design. Finally, using the interaction dependency network, modelling life cycle dependency aims to find cause-effect relationships among activities and determine key activities in sustainable design. If designers can find the right knowledge and understand dependency relations between interactions (activities), they will be able to find key design requirements to support sustainable design effectively.

A.1. Understanding a design problem

Understanding a design problem is the first step in eLCD, aiming to understand the natural language statement of a design problem through ROM as shown in Figure 17. ROM is a logic tool derived from the axiomatic theory of design modelling (ATDM) (Zeng 2002) for problem understanding, which is a graphic language to represent and reason about the meaning structure of the design problem through objects and their relationships (Zeng 2008).

ROM has two kinds of objects, namely, primitive and compound objects, and three kinds of relations: predicate, constraint, and connection, as shown in Table 6. Rules for generating ROM diagrams from the text can be found in (Zeng 2008). Examples of ROM diagrams are provided in Figure 8 to Figure 12.



Figure 17 Process of Understanding a Design Problem

Table 6 Elements of Recursive Object Model (ROM) (Zeng 2008, 2020b)

Type			Graphic Representation	Definition
Object	Primitive objects		E_i^a	Everything in the universe is an object.
	Object structure		$\oplus E$	It is an object that includes at least two other objects in it.
Relations	Connection		E_i — l → E_j	It is to connect two objects that do not constrain each other.
	Constraint		A ● → E_i	It is a descriptive, limiting, or particularizing relation of one object to another.
	Predicate	Subject-verb	S_i — v	It describes an act of an object on another or that describes the states of an object.
		Verb/proposition-object	v/p → O_j	

A.2. Decomposing life cycle

The decomposing life cycle is the second step in eLCD, which aims to collect the proper life cycle information through a question-driven approach, as shown in Figure 18. To collect the right information, one must ask the right type of question (one of 5W1H including what, who, why, when, where, and how) about the right object (topic) at the right time (order of questions) to the right resource. Section 2 frames the scope of the resources where the answers to the questions can be found, while this subsection will be focused on the order, topic, and type of the questions. Thus, the approach can be divided into two steps, which are question generation and question answering.



Figure 18 Process of Life Cycle Decomposition

Based on the ROM diagram generated in understanding a design problem, the right 5W1H questions can be generated following a question-asking algorithm. Then specific environment components and potential products can be analyzed from the three dimensions of the environment (environment types, life cycle events, and life cycle time) as shown in Figure 3. Finally, designers can merge the initial ROM diagram with the new information obtained from this question-driven process based on the collected information. Thus, the uncertainty of the design problem will be reduced.

A.2.1. Generating the right questions

A right question is the one asked at the right time about the right object with the right type and to the right resources (Zeng 2020b). Usually, 5W1H questions are asked in the design process based on designers' expertise and experience (Eris 2004). However, expert designers could tend to ask questions biased toward their expertise due to fixations, whereas novice designers could struggle to ask meaningful questions (Nguyen & Zeng 2017)**.** Design fixation refers to when designers use an inappropriate existing design idea to solve a design problem due to their strong attachment to the idea (Nguyen & Zeng 2017). Therefore, Wang and Zeng have proposed an algorithm to ask the right 5W1H questions to elicit the right product requirements (Wang & Zeng 2009), which is adopted in the eLCD method.

In Wang and Zeng's question-asking framework, two categories of questions are needed. The first aims to find out the real intent behind the design, simply asking a Why question for the design. The answer may lead to a reformulation of the design statement if the intent is different from the original design statement. The second focuses on meeting the identified real design intent by asking the right type (one of the 5W1H) of questions about the right object at the right time (order) to the right resource. The basic rules for the order to ask the second category of questions are shown in Table 7, whereas the type of questions for a selected object can be determined following the template in Table 8. Examples of such questions are provided in Table 1.

Table 7 Rules for the Right Order of Questions about the Right Objects (Wang & Zeng 2009)

Rule 1	Among all the candidate objects, the object with the most undefined constraints and predicates should be considered for questioning first.
Rule 2	Before an object can be asked for a question, the objects constraining or predicating them should be asked.

Table 8 Template for the Right Type of Questions (adapted from (Zeng 2020b))

Conditions		Question template
Noun (N)	Without any constraint	What/Who is N?
	With an adjective (A) constraint	What is A? What is A N?
Verb (V)	Subject (S_1) +Verb (V_1) +Object (O_1)	What do you mean S_1 V_1 O_1? Why do/does S_1 V_1 O_1? When do/does S_1 V_1 O_1? Where do/does S_1 V_1 O_1? How do/does S_1 V_1 O_1?

A.2.2. Collecting life cycle components

Collecting life cycle components aims to answer the questions asked from the previous step. Instead of answering the questions intuitively based on the designer's knowledge and experience, the answering template in Table 9 must be followed. While the questions based on Table 7 and Table 8 seek the necessary information for the design, the question-answering template provides a guideline to collect sufficient information related to the design. Founded on the three dimensions of environment structure, defined in Section 2, the template forces the collection of the information that would have significant influences on the potential product.

Examples of answers are provided in Table 2**.** Once the answers are collected**,** the original ROM diagram will be expanded by merging the ROM corresponding to each answer one by one. This merging process is illustrated in Figure 9 to Figure 12.

Table 9 eLCD-guided Question Answering (adapted from (Zeng 2020b))

Questions	Guideline
What/Who is N?	If N is the product to be designed or environment component of a product, then should add: [1] The purpose of N. [2] The life cycle of the N, and [3] The categorization of environment components for each event in the life cycle following the three dimensions of eLCD environment structure.

	If the N is an attribute of a product environment component, then define its relationship with the product or product environments.
Why do/does S_1 V_1 O_1?	To V_a O_a, S_1 V_1 O_1.
When do/does S_1 V_1 O_1?	In/On a time, S_1 V_1 O_1. When/During S_2 V_2 O_2, S_1 V_1 O_1.
Where do/does S_1 V_1 O_1?	In/Along a place, S_1 V_1 O_1. S_2 V_2 O_2, where S_1 V_1 O_1.
What do you mean by V_1? How do/does S_1 V_1 O_1?	If the S_1 and O_1 is not the product, the answer should include the life cycle of V_1. Else, skip the question and leave for solution synthesis.

A.2.3. Modelling life cycle dependency

The objective of the last eLCD step is to find key design requirements, as shown in Figure 19. Based on the collected life cycle information and the merged ROM diagram, key design requirements can be identified by analyzing interactions and their dependence relationships. Interactions are actions from the environment to the product and responses from the product to the environment (Zeng 2002).



Figure 19 Process of Modelling Life Cycle Dependency

The goal of this step is to identify basic messages from a design statement or situation. A basic message is an interaction, which is defined as a relation from one object to another that will generate a new object (Zeng & Gu 1999). An interaction is also a single activity that is created by one and only one action verb. Firstly, all the action verbs are extracted from the merged ROM diagram. Then, find the respective subjects and objects for all the extracted verbs. Finally, a verb, its subject, and object constitute an interaction.

The goal of this step is to find dependence relationships between interactions, which leads to an interaction dependence network. The interaction dependence network combines all interactions and their mutually dependent relations. A dependency relation of one interaction on the other refers to a situation where the absence of the other will lead to the absence of the first one, as shown in Figure 20. The determination of each dependency relation relies on the existing knowledge or can be conducted by searching for the knowledge relating to the two interactions. An example of an interaction dependence network is shown in Figure 13.

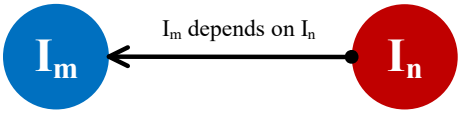


Figure 20 Dependency Relation between Two Interactions (Zeng 2020a)

The goal of this step is to find the independent and dependence-only interactions as well as the independent paths between them. Each path defines a sub-design problem, which is a major design activity implying key design requirements. Examples are shown in Table 5, Figure 14, and Figure 15.